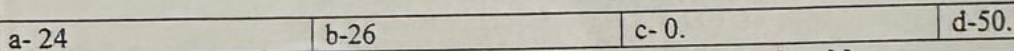


3. In PLCs rungs power flows from:

4. Timers may have one or more _____ for their reference timing.

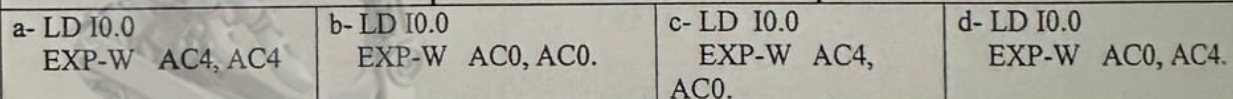
5. In an up-counter, when the accumulated count exceeds the preset count without a reset, the accumulated count will:

6. For the following ladder diagram suppose your accumulated value is 24 and you lose AC line power to the controller. When power is restored to your controller, the counter accumulated value will be:

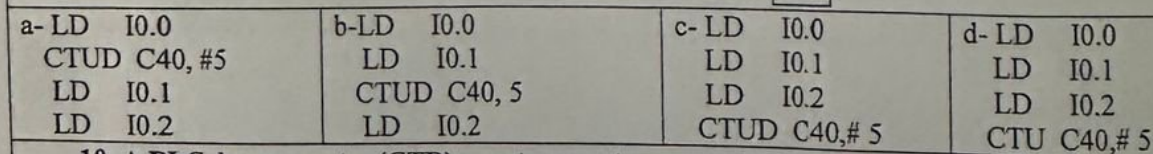


7. The length of time delay period in TOFF timer is calculated by:

8. The instruction list for the following ladder logic diagrams is:



9. The instruction list for the following ladder logic diagrams is:



10. A PLC down-counter (CTD) counter counts _____ transitions

- | | | | |
|-----------------|------------------|------------------|-------------------|
| a- true to true | b- false to true | c- false to zero | d- false to false |
|-----------------|------------------|------------------|-------------------|

[illegible]



Al-Balqa Applied University

جامعة البلقاء التطبيقية

Q4 (10 marks)

Write a ladder diagram program using Allen Bradley PLC unit to convert Celsius temperature to Fahrenheit based on the following equation $F^{\circ} = C^{\circ} \times \frac{9}{5} + 32$

The program is designed to convert the recorded Celsius temperature in the data table to Fahrenheit values for display (O:13).

